**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 005 · Images, Pixels, Colour Depth and File Sizes

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is a character set?

Q2. Using decimal units, how many KB are in 3 MB?
A. 3,000 KB
B. 3 KB
C. 300 KB
D. 30,000 KB

Q3. Use decimal units. Convert 3 TB to GB, then state how many TB are in 1 PB.

Q4. Two computers receive the same byte but show different characters. Why? What must match?

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# I Do · Read new knowledge

## Key vocabulary

pixel — One individual picture element of a bitmap.

resolution — The number of pixels across and down an image.

colour depth — Bits used to store a pixel's colour.

metadata — Additional information stored about a file or image.

## New learning

Bitmap images: A bitmap is made of pixels. Resolution describes its width and height in pixels.

Colour depth: Colour depth is bits allocated to each pixel. An n-bit colour depth can represent up to 2^n different colours.

Raw file-size method: Multiply width × height × colour depth to obtain bits. Divide by 8 to convert to bytes.

Assumptions: This calculates uncompressed pixel data only. File headers, metadata and compression may change the actual file size.

## Worked example

Calculate raw storage for an 8 × 8 bitmap at 2-bit colour depth.

1. Resolution is 8 × 8 = 64 pixels.

2. Each pixel uses 2 bits, so 64 × 2 = 128 bits.

3. Convert to bytes: 128 ÷ 8 = 16 bytes.

4. State the unit and the assumption: no headers or compression.

# We Do · Practise with guidance

### Guided 1

A bitmap is 10 × 6 pixels. How many pixels are there?

Worked method: 60 pixels.

### Guided 2

At 4-bit colour depth, how many bits for that image?

Worked method: 60 × 4 = 240 bits.

### Guided 3

Convert the 240 bits to bytes and explain what happens if depth doubles.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: 240 ÷ 8 = 30 bytes; doubling depth doubles raw pixel-data size to 60 bytes.

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# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Write one print instruction to calculate the size in bits.

```
width = 10
height = 8
depth = 4
```

write

### Question 1

State what is meant by a pixel and colour depth. [2 marks]

### Question 2

A bitmap is 12 × 5 pixels at 4-bit colour depth. Calculate raw bits and bytes. [3 marks]

### Question 3

A 1,000 × 800 bitmap has 8-bit colour depth. Calculate raw storage in MB using 1,000 bytes per KB and 1,000 KB per MB. [4 marks]

# Exit Ticket · Answer before checking

Doubling colour depth doubles the uncompressed pixel-data size if dimensions stay the same.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A defined collection of characters with corresponding codes.

Q2. A — 3,000 KB

Q3. 3,000 GB; 1 PB = 1,000 TB.

Q4. The computers use different character sets or encodings. To show the same letter, both need a matching encoding.

### I Do / We Do pages · Guided practice

Q1. 60 pixels.

Q2. 60 × 4 = 240 bits.

Q3. 240 ÷ 8 = 30 bytes; doubling depth doubles raw pixel-data size to 60 bytes.

### You Do pages · Code First

```
width = 10
height = 8
depth = 4
print(width * height * depth)
```

Expected result: 320

### You Do pages · Independent questions

Q1. Pixel: smallest individual element in a bitmap [1]; colour depth: bits used to represent each pixel's colour [1].

Q2. 12 × 5 = 60 pixels [1]; 60 × 4 = 240 bits [1]; 240 ÷ 8 = 30 bytes [1].

Q3. 1,000 × 800 = 800,000 pixels [1]; × 8 = 6,400,000 bits [1]; ÷8 = 800,000 bytes [1]; ÷1,000,000 = 0.8 MB [1].

### Exit Ticket · Model answer

True

## Try the question again

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**